
TV-Bro Architecture - Layered View
7. Service/Application
TVBro
(Application, wiring)
DownloadService
DownloadTask
host (implements WebEngineCallbackHost)
6. Engine
WebEngineCallback
(per-tab adapter, injected models)
GeckoWebEngine
WebViewWebEngine
WebEngineWindowProviderCallback
(engine<->app)
WebEngine
(interface)
ActiveModelsRepository
(lifecycle-scoped holders)
observable state
observable
observable
observable
observable
observable
observable
observable state
observable
observable
1. UI Layer
ActionBar/View Layer
(TabsView, BookmarksPanelView, ActionBar)
SettingsDialog/Pages
(declarative SettingsItem)
DownloadsActivity
(active + history + actions)
HistoryActivity
(list + search + multi-select)
MainActivity
(implements WebEngineCallbackHost
+ lifecycle + wiring)
PanelController
(side panels + row focus)
ToolbarController
(overlay + fullscreen)
PermissionOrchestrator
(permission/result routing)
MediaListPresenter
(stream list dialog)
IntentHandler
(deep links + history result)
3. Domain (pure, framework-free)
NewTabPositionPolicy
(new tab insertion index)
SiteDomains
(same-site/domain checks)
NavigationPolicy
(scheme routing + redirect blocking)
PopupPolicy
(popup blocking rules)
AddressInputPolicy
(URL/search resolution)
UrlNormalizer
(URL normalization)
LinkActionPolicy
(link action decisions)
2. Presentation Models (ActiveModel)
TabsModel
(tabs state, current tab,
open/close/switch)
AdblockModel
(adblock client, rules)
MainActivityViewModel
(homepage links, bookmarks,
history log, incognito cleanup)
SettingsModel
(config-backed observables,
user prefs)
AutoUpdateModel
(update checks/state)
DownloadsHistoryModel
(paginated history)
ActiveDownloadsModel
(active tasks + listeners)
HistoryModel
(paginated search/list)
4. Repository (suspend-only)
Repositories
(manual DI container, lazy)
FavoritesRepository
HistoryRepository
HostsRepository
TabsRepository
DownloadsRepository
5. Data (Room)
AppDatabase
FavoritesDao
HistoryDao
HostsDao
TabsDao
DownloadDao